Practical Tutorial: Custom Skill Creation

Building .agents/skills/gemma-eval/SKILL.md with Explicit YAML Frontmatter & Markdown Body

Step-by-Step Overview: *This guide walks through the exact process of authoring a workspace custom skill in Google Antigravity. A skill comprises two critical components: (1)* ***YAML Frontmatter*** *defining the trigger contract used for discovery via progressive disclosure, and (2)* ***Markdown Body*** *containing actionable runbooks, helper scripts, and validation criteria.*

Step 1: Create the Skill Directory Layout

Inside your project root, create the standard directory structure under .agents/skills/gemma-eval/:

```
# In Terminal (PowerShell or Bash) at Project Root:
mkdir -p .agents/skills/gemma-eval/scripts
mkdir -p .agents/skills/gemma-eval/references
```

Step 2: Author the YAML Frontmatter (The Trigger Contract)

Open .agents/skills/gemma-eval/SKILL.md. The file **MUST** begin on Line 1 with three dashes (---) and conclude the frontmatter with three dashes (---).

```
---
name: gemma-eval
description: >-
  Use this skill when the user asks to evaluate, benchmark, fine-tune, or test
  Gemma models (e.g., Gemma 2B, 7B, 9B, 27B), run automated model evaluations,
  compute perplexity/loss metrics, or validate inference prompt templates.
---
```

Frontmatter Key Guidelines:

- **name:** Must match the directory name (gemma-eval), lowercase and hyphenated.
- **description:** Written in third-person starting with *'Use this skill when...'*. List specific target keywords, domain concepts, model variants, and command intents.

Step 3: Author the Markdown Body & Procedural Steps

Below the frontmatter, write the actionable Markdown runbook. Use clear numbered stages, relative links to helper scripts, and automated verification commands:

````
# Gemma Model Evaluation & Benchmarking Runbook

This runbook guides the agent through pre-flight checks, benchmark execution, and metric analysis for Gemma checkpoi
nts.

## 1. Environment Pre-Flight Check
Before starting, execute the pre-flight verification script to check GPU availability and dependencies:
- [verify_environment.py](./scripts/verify_environment.py)

## 2. Automated Benchmark Execution
Run the evaluation harness specifying model checkpoint, benchmark dataset, and batch configuration:
```bash
python .agents/skills/gemma-eval/scripts/evaluate_model.py \
    --model-id "google/gemma-2-9b-it" \
    --benchmark "mmlu-pro" \
    --batch-size 4 \
    --output-dir "./eval_results"
```

## 3. Metric Inspection & Validation
Inspect generated metrics in `./eval_results/metrics.json` against standard baseline thresholds:
- [metrics_reference.md](./references/metrics_reference.md)

## 4. Verification Criteria
1. Confirm `./eval_results/metrics.json` and `./eval_results/eval_summary.md` are present.
2. Ensure perplexity score is <= 8.5 and MMLU-Pro accuracy is >= 65.0%.
3. Confirm zero CUDA out-of-memory (OOM) exceptions in log output.
````

Step 4: Add Helper Scripts & Reference Manuals

Leverage **Progressive Disclosure** by isolating complex logic and extensive documentation into modular files:

- **scripts/verify_environment.py:** Python script checking PyTorch version, CUDA devices, and Hugging Face Transformers installation.
- **references/metrics_reference.md:** Reference tables containing baseline scores (MMLU, GSM8K, HumanEval) and troubleshooting procedures.

Step 5: Testing & Validating Skill Discovery

Once files are saved, Antigravity automatically indexes the skill upon your next message. You can test it by prompting:

> *'Evaluate the Gemma 2 9B model on MMLU-Pro and check perplexity.'*

The agent will match the trigger description, dynamically load SKILL.md, inspect verify_environment.py, and execute the runbook.